Physical Chemistry (I)

Lecture 21

Colligative Properties

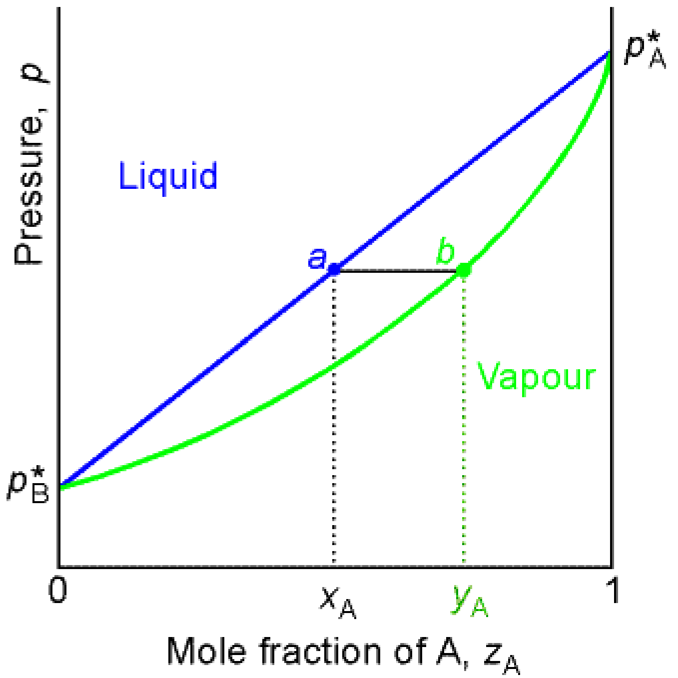
Chemical Reactions

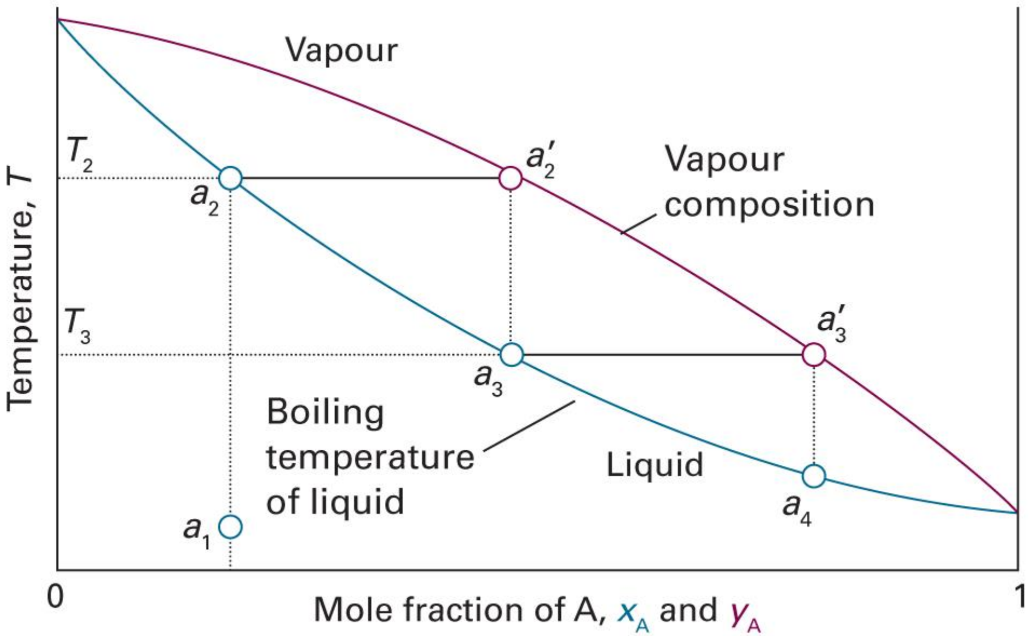
In the Last Class

- Phase diagrams of ideal solutions
- Phase diagrams of real solutions
  - Azeotropes
  - Phase separation
- Phase diagrams of liquid-solid mixtures

Last class

Ideal Solutions





(Images: http://people.cst.cmich.edu/teckl1mm/pchemi/chm351ch8af01.htm
Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.4)

Real Solutions

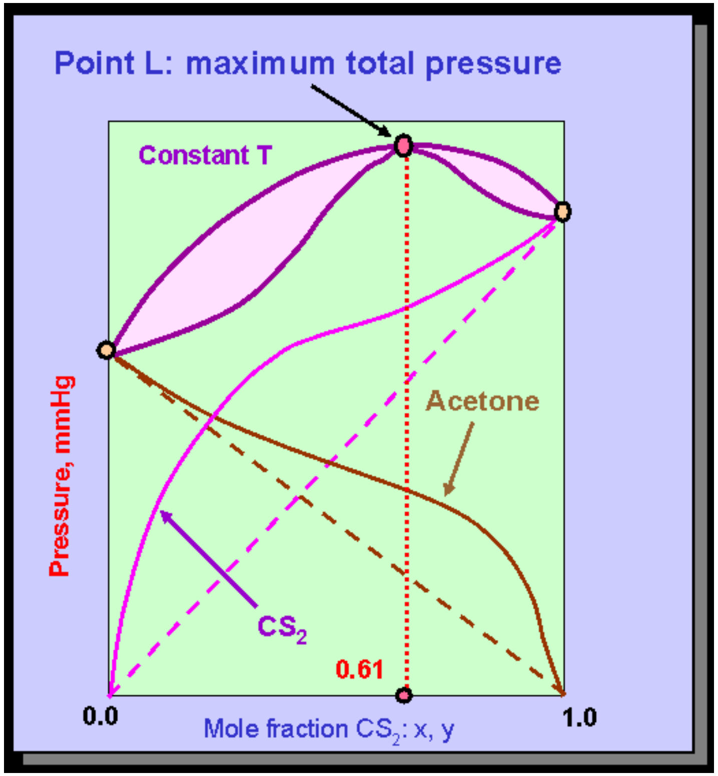
$$\xi = \frac{z}{k_\mathrm{B}T}\left(w_\mathrm{AB} - \frac{w_\mathrm{AA} + w_\mathrm{BB}}{2}\right) \neq 0 \quad \text{(BW model)}$$

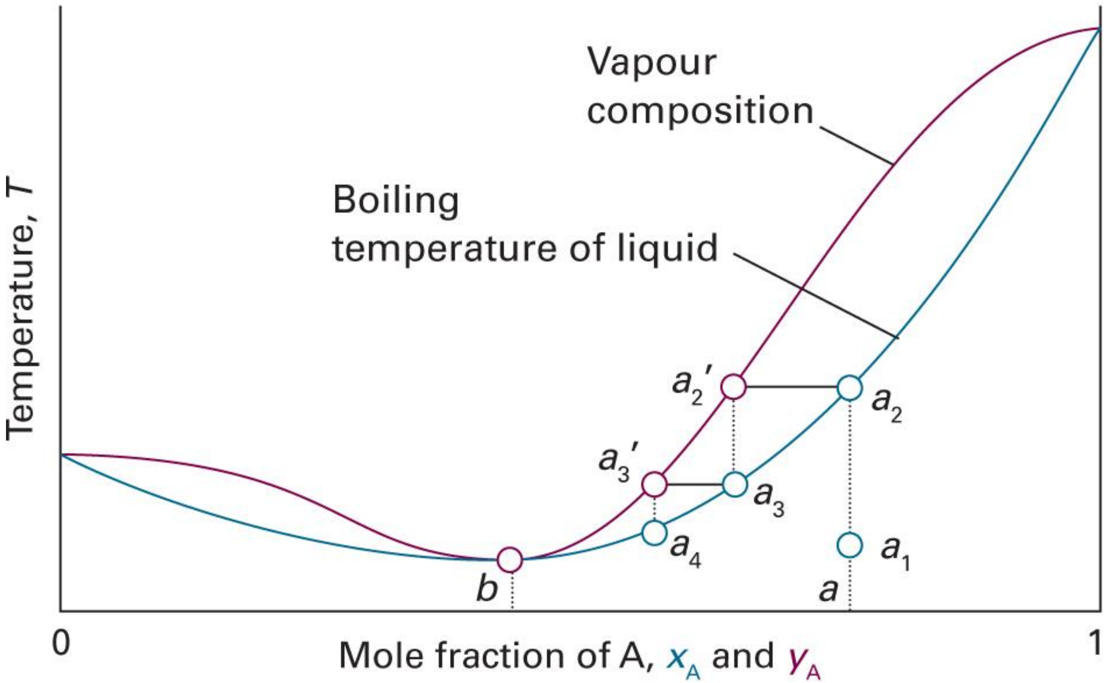
- $\xi > 0$: AB interactions are weaker
- $\xi < 0$: AB interactions are stronger

1. Change in liquid-gas equilibrium: azeotrope
2. Change in liquid-liquid equilibrium: phase separation

Last class

For $\xi > 0$

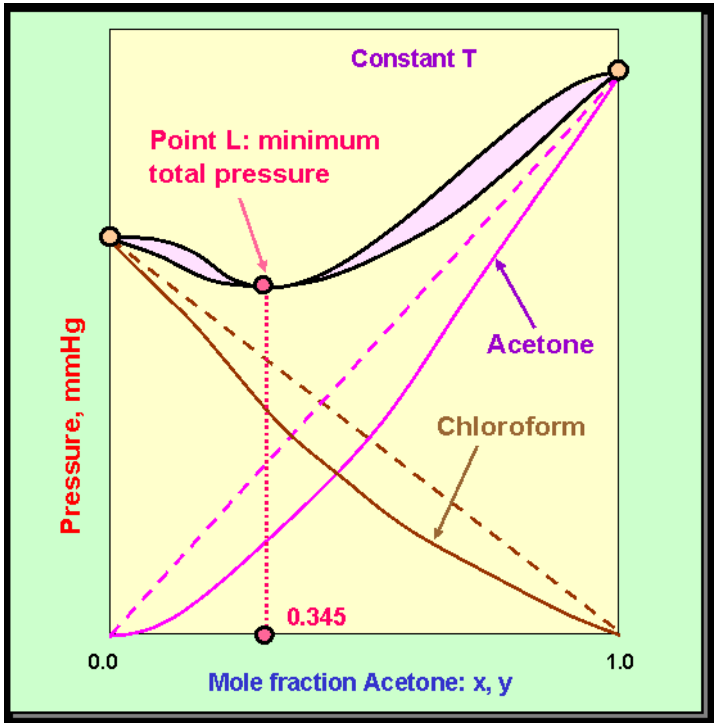


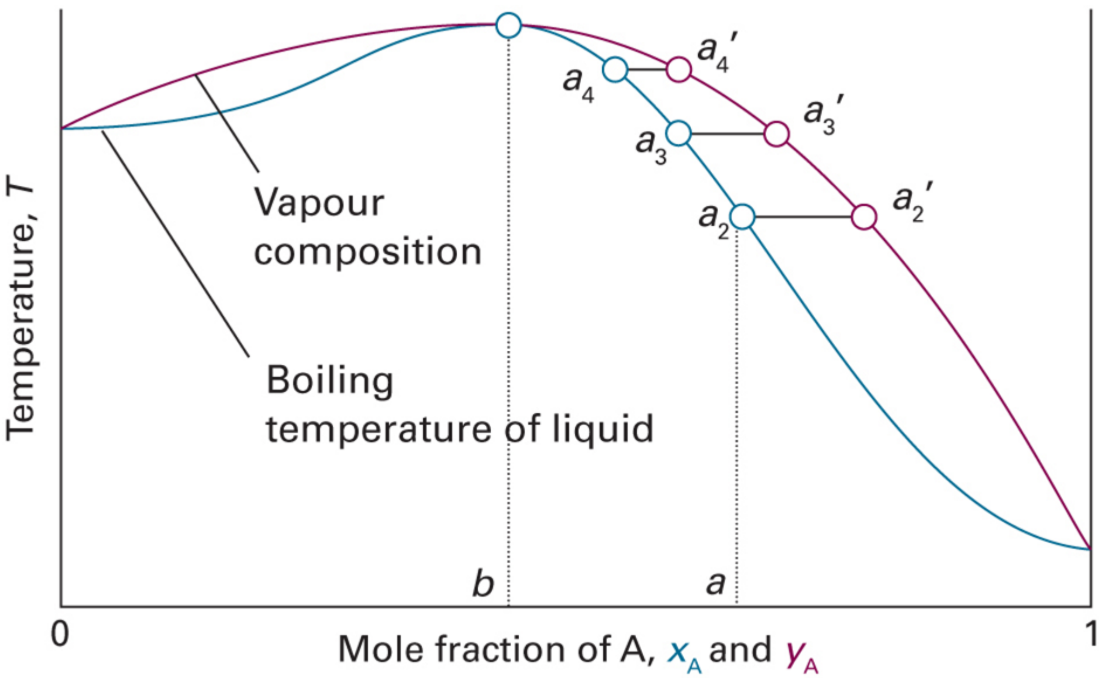


(Images: http://www.separationprocesses.com/Distillation/DT_Chp01f.htm
Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.11)

Last class

For $\xi < 0$

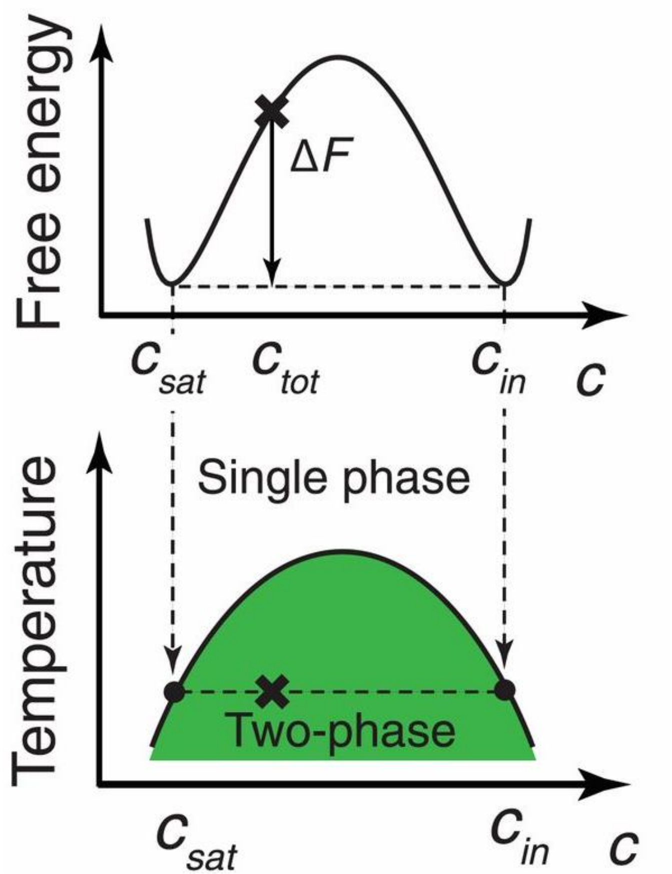




(Images: http://www.separationprocesses.com/Distillation/DT_Chp01f.htm
Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.10)

Last class

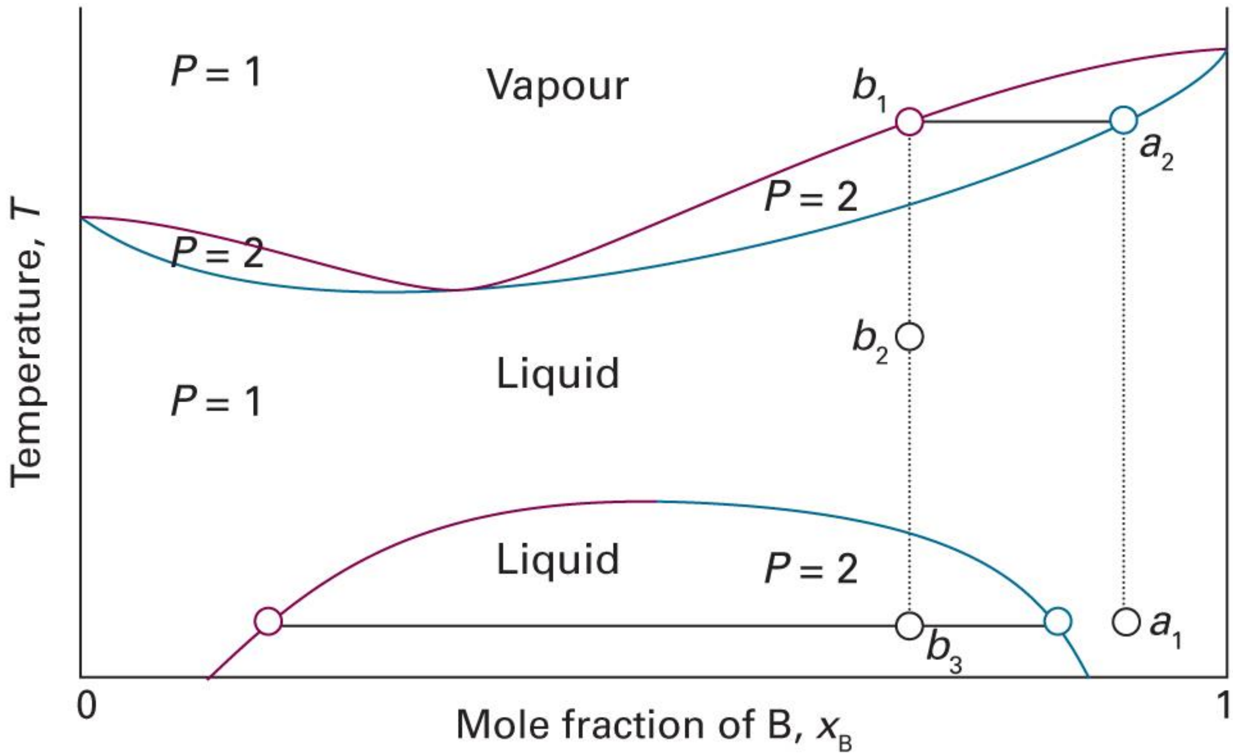
Phase Separation at $\xi > 2$



(Images: Shin and Brangwynne, *Science* 357: eaaf4382 (2017), Figure 3
https://commons.wikimedia.org/wiki/File:Olive_oil_with_Balsamic_Vinegar.jpg)

Last class

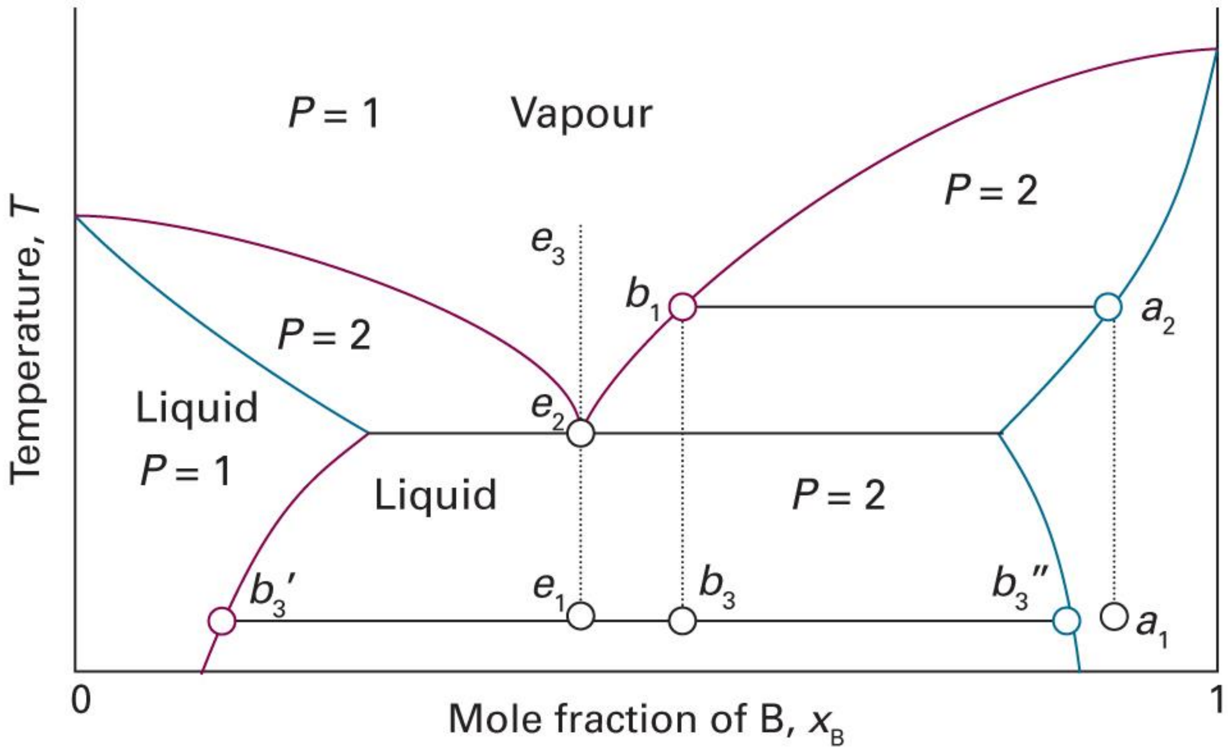
Miscible before Boiling



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.20)

Last class

Boiling before Miscible

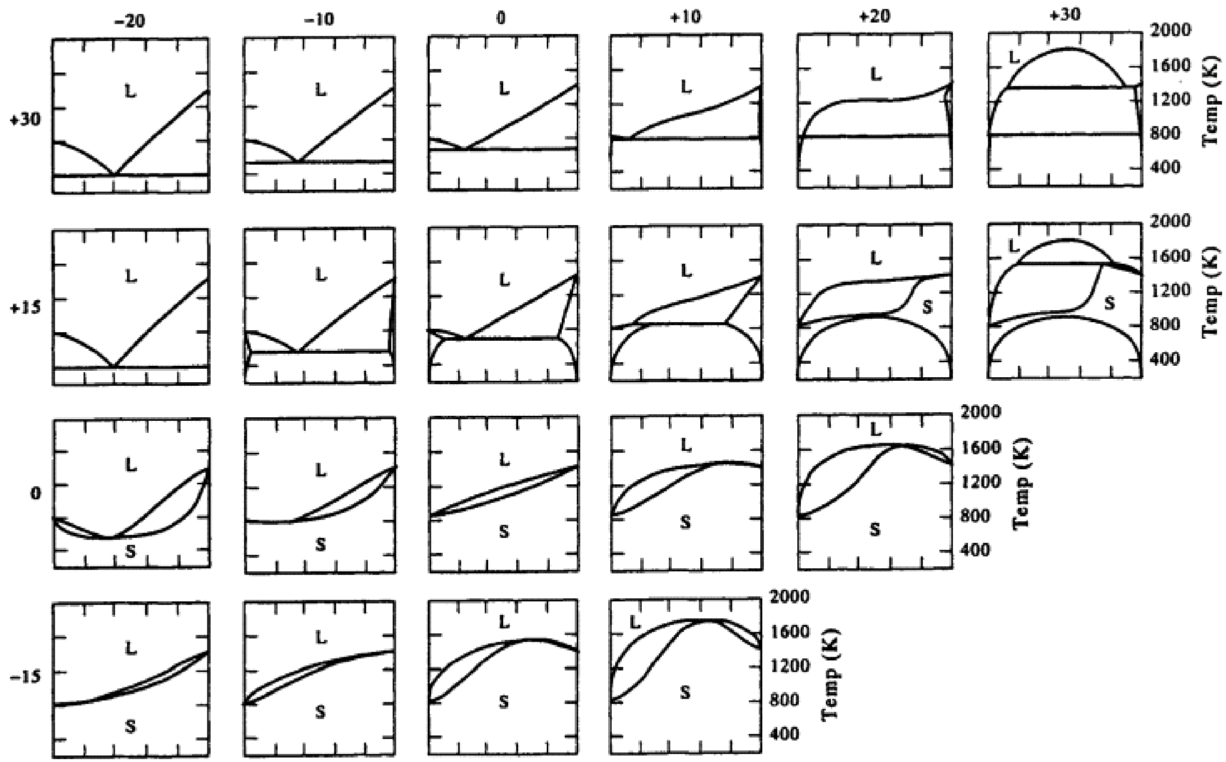


(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.21)

Solid-Liquid Equilibrium

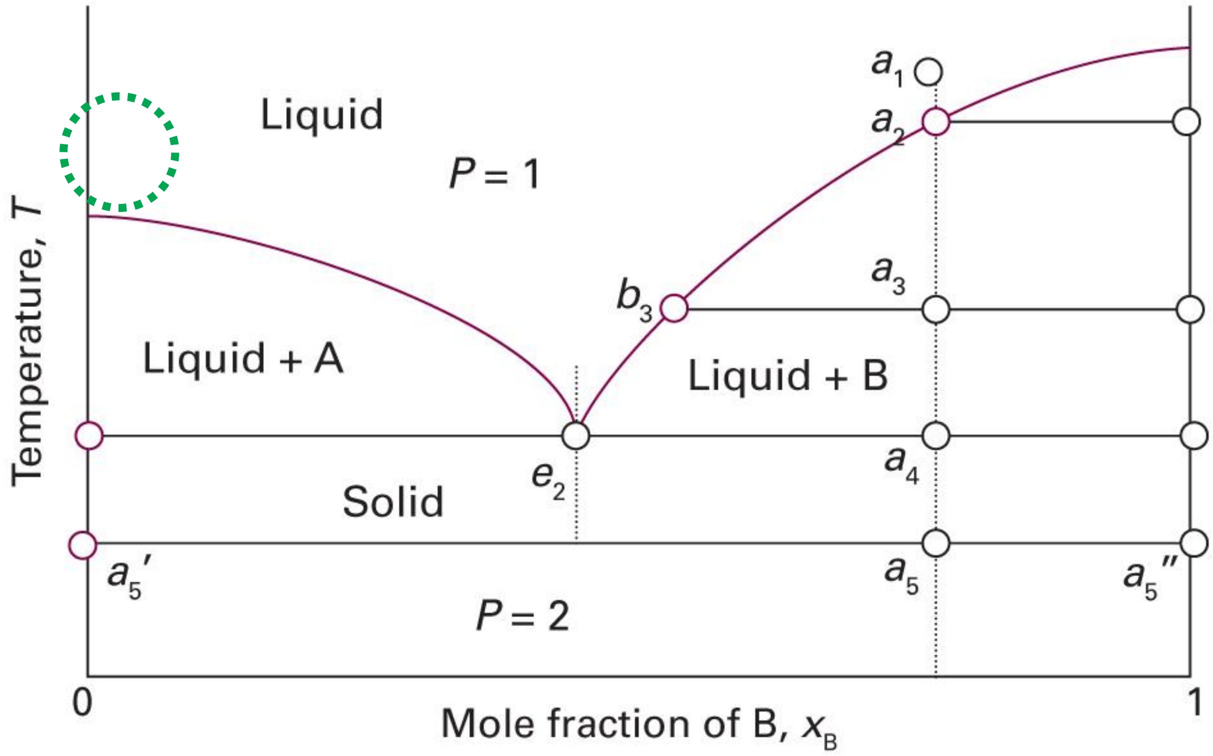
ξ_{liq}

ξ_{sol}



(Image: Gaskell, *Introduction to the Thermodynamics of Materials*, 5th Ed., Fig. 10.23)

A Little Solid in Liquid



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5D.1)

A Little Solid in Liquid

- Considered as an ideal-dilute solution
  - Liquid: solvent
  - Solid: solute

- Two assumptions
  - The solute is not volatile
  - The solute does not dissolve in the solid solvent

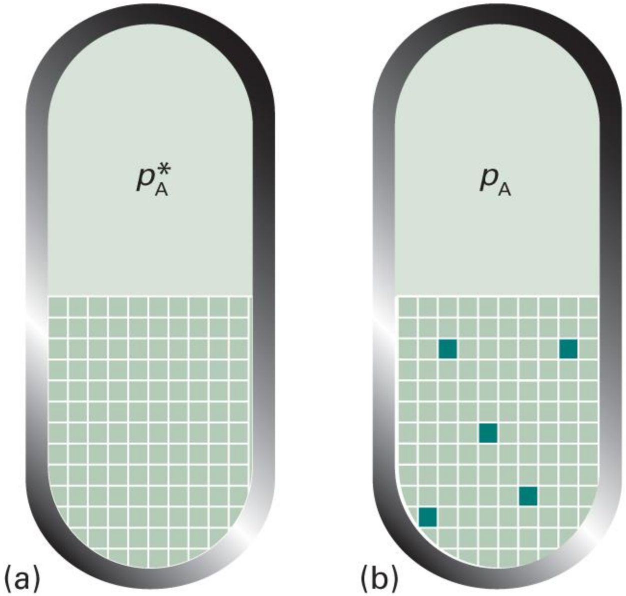
Colligative Properties

- Physical properties that depends on the relative number of solute particles present but not their chemical identity

1. Lowering of vapor pressure
2. Elevation of boiling point
3. Depression of freezing point
4. Osmotic pressure

Origin of Colligative Properties

- Due to the reduction of the chemical potential of the liquid solvent as a result of the presence of solute



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5B.7)

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Lowering of Vapor Pressure

From Raoult's law,

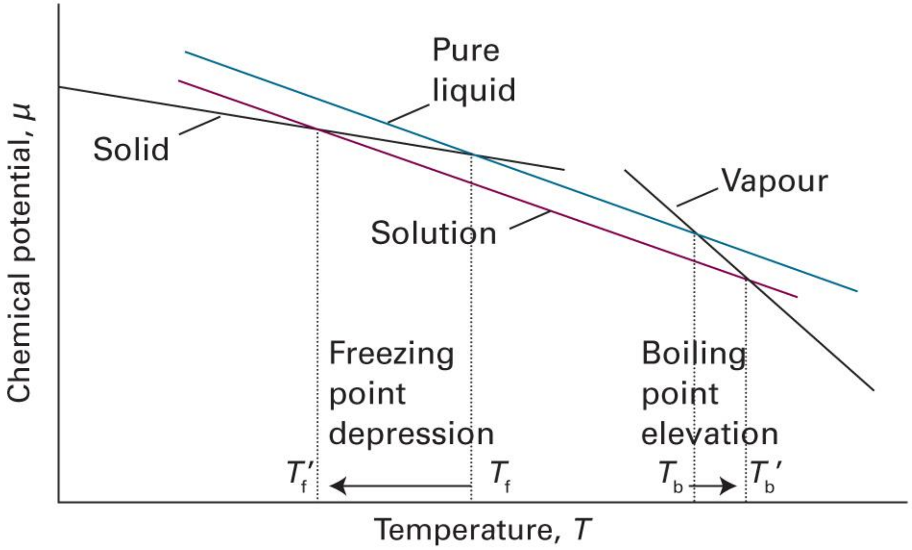
$$p_{\mathrm{A}} = x_{\mathrm{A}} p_{\mathrm{A}}^{*}$$

The change in vapor pressure is

$$\Delta p_{\mathrm{A}} = p_{\mathrm{A}}^{*} - p_{\mathrm{A}} = (1 - x_{\mathrm{A}}) p_{\mathrm{A}}^{*} = x_{\mathrm{B}} p_{\mathrm{A}}^{*}$$

Changes in Phase Transitions

$$\mu_A = \mu_A^* + RT \ln x_A \; (x_A < 1)$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5B.6)

Boiling: Liquid-Gas Equilibrium

$$\mu_A^*(g) = \mu_A(l)$$

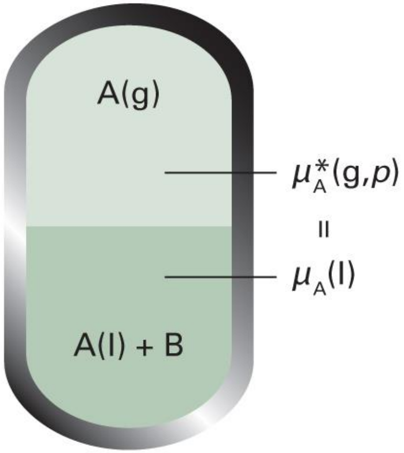
$$\mu_A^*(g) = \mu_A^*(l) + RT \ln x_A$$

which can be rearranged as

$$\ln x_A = \frac{\mu_A^*(g) - \mu_A^*(l)}{RT} = \frac{\Delta_{\text{vap}} G}{RT}$$

Using the Gibbs-Helmholtz equation,

$$\frac{\mathrm{d} \ln x_A}{\mathrm{d}T} = \frac{1}{R} \frac{\mathrm{d}\left(\Delta_{\text{vap}} G/T\right)}{\mathrm{d}T} = -\frac{\Delta_{\text{vap}} H}{RT^2}$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5B.8)

Integration

$$\frac{\mathrm{d}\ln x_\mathrm{A}}{\mathrm{d}T} = -\frac{\Delta_\mathrm{vap}H}{RT^2}$$

$$\mathrm{d}\ln x_\mathrm{A} = -\frac{\Delta_\mathrm{vap}H}{RT^2}\mathrm{d}T$$

$$\int_\text{pure solvent}^\text{current state} \mathrm{d}\ln x_\mathrm{A} = -\frac{1}{R}\int_\text{pure solvent}^\text{current state} \frac{\Delta_\mathrm{vap}H}{T^2}\mathrm{d}T$$

$$\int_0^{\ln x_\mathrm{A}} \mathrm{d}\ln x'_\mathrm{A} = -\frac{1}{R}\int_{T^*}^{T} \frac{\Delta_\mathrm{vap}H}{(T')^2}\mathrm{d}T'$$

Evaluation

$$\int_0^{\ln x_A} \mathrm{d}\ln x_A' = -\frac{1}{R}\int_{T^*}^{T} \frac{\Delta_{\text{vap}}H}{(T')^2}\,\mathrm{d}T'$$

Assuming that $\Delta_{\text{vap}}H$ is constant,

$$\ln x_A = -\frac{\Delta_{\text{vap}}H}{R}\int_{T^*}^{T}\frac{1}{(T')^2}\,\mathrm{d}T' = -\frac{\Delta_{\text{vap}}H}{R}\left(-\frac{1}{T}+\frac{1}{T^*}\right)$$

$$\ln(1-x_B) = \frac{\Delta_{\text{vap}}H}{R}\left(\frac{1}{T}-\frac{1}{T^*}\right)$$

Two (Valid) Approximations

$$\ln(1 - x_\mathrm{B}) = \frac{\Delta_\mathrm{vap}H}{R}\left(\frac{1}{T} - \frac{1}{T^*}\right)$$

For $x_\mathrm{B} \ll 1$, $\ln(1 - x_\mathrm{B}) \approx -x_\mathrm{B}$ and

$$x_\mathrm{B} \approx \frac{\Delta_\mathrm{vap}H}{R}\left(\frac{1}{T^*} - \frac{1}{T}\right) = \frac{\Delta_\mathrm{vap}H}{RTT^*}(T - T^*)$$

Also, supposing $\Delta T_\mathrm{b} = T - T^* \ll T^*$,

$$x_\mathrm{B} \approx \frac{\Delta_\mathrm{vap}H}{R(T^*)^2}\Delta T_\mathrm{b}$$

Elevation of Boiling Point

$$x_{\mathrm{B}} \approx \frac{\Delta_{\mathrm{vap}}H}{R(T^*)^2}\Delta T_{\mathrm{b}}$$

Rearrangement gives

$$\Delta T_{\mathrm{b}} = Kx_{\mathrm{B}}$$

where $K = R(T^*)^2/\Delta_{\mathrm{vap}}H$.

Using the molality,

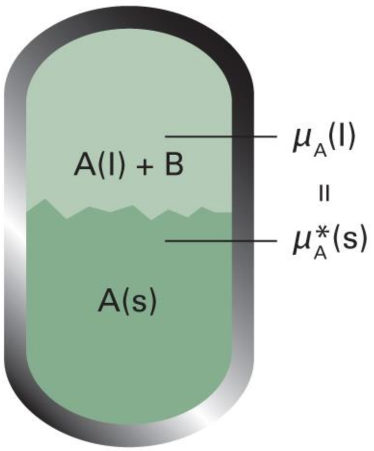
$$\Delta T_{\mathrm{b}} = K_{\mathrm{b}}b$$

where K_{b} is called the boiling-point constant.

Freezing: Solid-Liquid Equilibrium

$$\mu_A^*(s) = \mu_A(l)$$

$$\mu_A^*(s) = \mu_A^*(l) + RT \ln x_A$$



Similarly, we obtain

$$\Delta T_f = K' x_B$$

where $K' = R(T^*)^2/\Delta_{fus}H$, or,

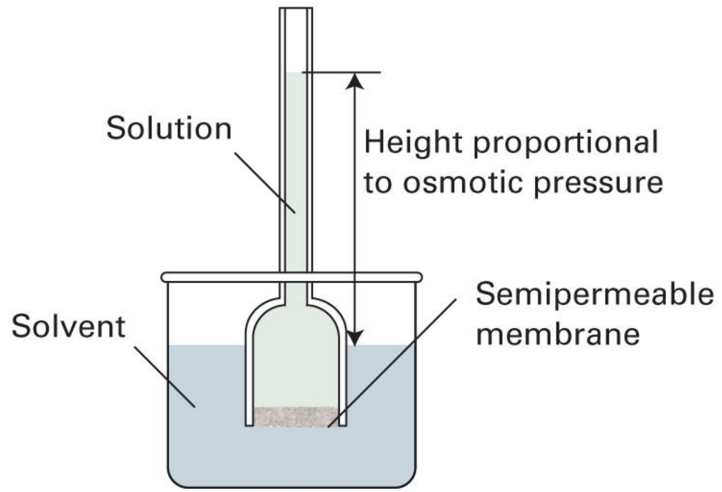
$$\Delta T_f = K_f b$$

where K_f is called the freezing-point constant.

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5B.9)

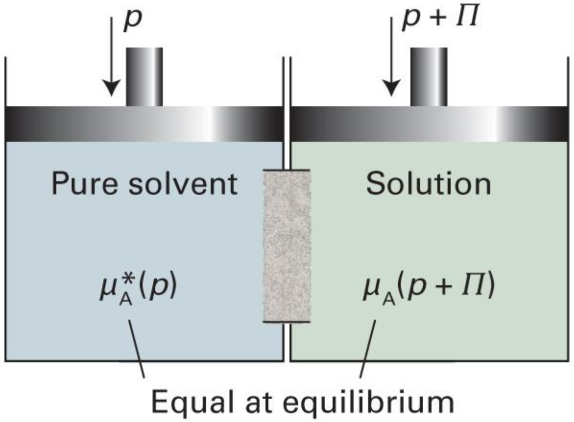
Osmosis

the spontaneous passage of a pure solvent into a solution separated from it by a semipermeable membrane



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5B.13)

Thermodynamics of Osmosis



$$\mu_{\mathrm{A}}^*(p) = \mu_{\mathrm{A}}(x_{\mathrm{A}}, p + \Pi)$$

$$\mu_{\mathrm{A}}^*(p) = \mu_{\mathrm{A}}^*(p + \Pi) + RT \ln x_{\mathrm{A}}$$

Thus,

$$\mu_{\mathrm{A}}^*(p + \Pi) = \mu_{\mathrm{A}}^*(p) - RT \ln x_{\mathrm{A}}$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5B.12)

Pressure Effect

In general,

$$G_{\mathrm{m}}(p_2) = G_{\mathrm{m}}(p_1) + \int_{p_1}^{p_2} V_{\mathrm{m}}\,\mathrm{d}p$$

$$\mu_{\mathrm{A}}^{*}(p+\Pi) = \mu_{\mathrm{A}}^{*}(p) + \int_{p}^{p+\Pi} V_{\mathrm{m}}\,\mathrm{d}p'$$

Assuming that V_{m} is constant,

$$\mu_{\mathrm{A}}^{*}(p+\Pi) = \mu_{\mathrm{A}}^{*}(p) + V_{\mathrm{m}}\int_{p}^{p+\Pi} 1\,\mathrm{d}p' = \mu_{\mathrm{A}}^{*}(p) + V_{\mathrm{m}}\Pi$$

Combining the Two

$$\mu_{\mathrm{A}}^{*}(p+\Pi)=\mu_{\mathrm{A}}^{*}(p)-RT\ln x_{\mathrm{A}}$$

$$\mu_{\mathrm{A}}^{*}(p+\Pi)=\mu_{\mathrm{A}}^{*}(p)+V_{\mathrm{m}}\Pi$$

Comparing the two equations,

$$-RT\ln x_{\mathrm{A}}=V_{\mathrm{m}}\Pi$$

$$-RT\ln(1-x_{\mathrm{B}})=V_{\mathrm{m}}\Pi$$

Since $\ln(1-x_{\mathrm{B}})\approx -x_{\mathrm{B}}$ for $x_{\mathrm{B}}\ll 1$,

$$RTx_{\mathrm{B}}=V_{\mathrm{m}}\Pi$$

Osmotic Pressure

$$RTx_{\mathrm{B}} = V_{\mathrm{m}}\Pi$$

For a dilute solution,

$$x_{\mathrm{B}} = n_{\mathrm{B}}/(n_{\mathrm{A}} + n_{\mathrm{B}}) \approx n_{\mathrm{B}}/n_{\mathrm{A}}$$

Thus,

$$RTn_{\mathrm{B}} = n_{\mathrm{A}}V_{\mathrm{m}}\Pi = V\Pi$$

and

$\Pi = [\mathrm{B}]RT$ (van't Hoff equation)

Real Solutions

For an ideal solution (now solute = J),

$$\Pi = [\mathrm{J}]RT$$

The osmotic virial expansion is

$$\Pi = [\mathrm{J}]RT(1 + B[\mathrm{J}] + C[\mathrm{J}]^2 + \cdots)$$

where $B, C, \ldots$ are the osmotic virial coefficients.

Second Coefficient

$$\Pi = [\mathrm{J}]RT(1 + B[\mathrm{J}] + \cdots)$$

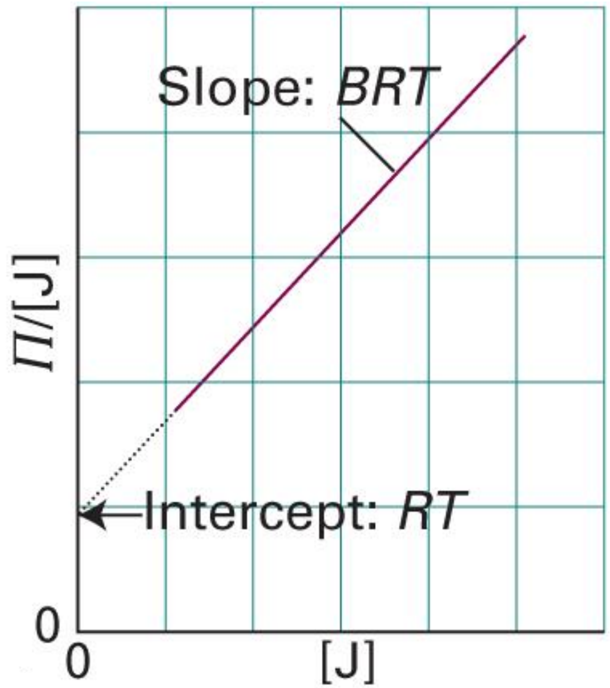
Taking the first correction term only,

$$\Pi \approx [\mathrm{J}]RT(1 + B[\mathrm{J}])$$

- $B > 0$: solute-solute repulsion dominant
- $B \approx 0$: similar to ideal-dilute solution
- $B < 0$: solute-solute attraction dominant

Osmometry

$$\Pi/[\mathrm{J}] \approx RT + BRT[\mathrm{J}]$$



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Case of Ionic Solutes

e.g. $NaCl(s) \longrightarrow Na^{+}(aq) + Cl^{-}(aq)$

- van’t Hoff factor i

$$\Delta T_{\mathrm{b}} = iK_{\mathrm{b}}b$$

$$\Delta T_{\mathrm{f}} = iK_{\mathrm{f}}b$$

$$\Pi = i[\mathrm{B}]RT$$

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van't Hoff Factors

0.05000 M, 25℃

Compound	i (ideal)	i (measured)
glucose	1	1.0
NaCl	2	1.9
KCl	2	1.9
$MgCl_2$	3	2.7
$FeCl_3$	4	3.4
$Ca(NO_3)_2$	3	2.5
$AlCl_3$	4	3.2
$MgSO_4$	2	1.4

(Image: https://chem.libretexts.org/@go/page/24261)

Summary

- Phase
- Chemical potential
- Gibbs-Duhem eq.

Multi-component system

Gas mixture

- Dalton's law
- Perfect gas
- Real gas

Liquid mixture

Ideal solution

- Raoult's law

$$p_A = x_A p_A^*$$

- Chemical potential

$$\mu_A = \mu_A^* + RT \ln x_A$$

- $\Delta_{mix} G = -T \Delta_{mix} S$

Real solution

- Regular solution
- Bragg-Williams model
- Excess function
- Activity

$$\mu_A = \mu_A^* + RT \ln a_A$$

$$\mu_B = \mu_B^{\ominus} + RT \ln a_B$$

Ideal-dilute solution

- Henry's law

$$p_B = x_B K_B$$

- Chemical potential

$$\mu_B = \mu_B^{\ominus} + RT \ln x_B$$

- L-G phase diagram
- L-L phase diagram
- L-S phase diagram

Colligative properties

9. Chemical Equilibria

Chapter Overview

- Chemical reactions and equilibria
  - Reaction quotient and equilibrium constant
  - Activity approximations
- Le Chatelier's principle
  - Amount change
  - Volume change
  - Temperature change

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Chemical Reactions

$$aA + bB \rightleftarrows cC + dD$$

Can we construct

thermodynamics of chemical reactions?

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Extent of Reaction ξ

$$a\mathrm{A} + b\mathrm{B} \rightleftarrows c\mathrm{C} + d\mathrm{D}$$

Species	Initial	→	Change	→	Final
A	n_A		$-a\xi$		$n_\mathrm{A} - a\xi$
B	n_B		$-b\xi$		$n_\mathrm{B} - b\xi$
C	n_C		$+c\xi$		$n_\mathrm{C} + c\xi$
D	n_D		$+d\xi$		$n_\mathrm{D} + d\xi$

Simple Reaction

$$A(g) \rightleftarrows B(g)$$

- Extent of reaction ξ: for $\mathrm{d}\xi$, $\mathrm{d}n_\mathrm{A} = -\mathrm{d}\xi$ and $\mathrm{d}n_\mathrm{B} = +\mathrm{d}\xi$
- Gibbs energy $G = G(p, T, n_\mathrm{A}, n_\mathrm{B})$

  For constant p and T,

$$\mathrm{d}G = \mu_\mathrm{A}\mathrm{d}n_\mathrm{A} + \mu_\mathrm{B}\mathrm{d}n_\mathrm{B} = -\mu_\mathrm{A}\mathrm{d}\xi + \mu_\mathrm{B}\mathrm{d}\xi = (\mu_\mathrm{B} - \mu_\mathrm{A})\mathrm{d}\xi$$

$$\left(\frac{\partial G}{\partial \xi}\right)_{p,T} = \mu_\mathrm{B} - \mu_\mathrm{A}$$

Reaction Gibbs Energy

$$A(g) \rightleftarrows B(g)$$

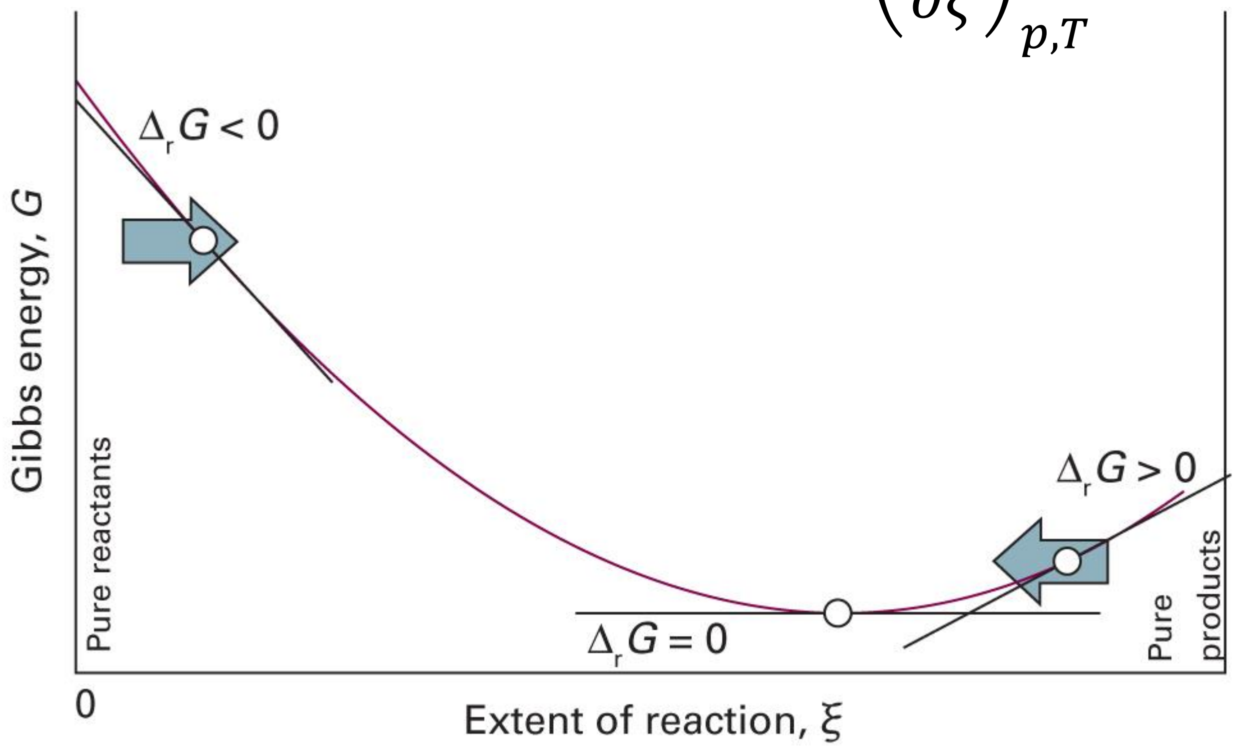
Assuming the absolute scale,

$$\Delta_r G = G_{m,B} - G_{m,A} = \mu_B - \mu_A = \left(\frac{\partial G}{\partial \xi}\right)_{p,T}$$

- $\Delta_r G = 0$: reversible, equilibrium
- $\Delta_r G < 0$: irreversible, spontaneous reaction
- $\Delta_r G > 0$: impossible $\rightarrow$ spontaneous reverse reaction

Progress of Reaction

$$G = \mu_\mathrm{A} n_\mathrm{A} + \mu_\mathrm{B} n_\mathrm{B} \qquad \Delta_\mathrm{r} G = \left(\frac{\partial G}{\partial \xi}\right)_{p,T}$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 6A.1)

Reaction Quotient

A(g) ⇄ B(g)

For a perfect gas, $\mu = \mu^{\ominus} + RT\ln(p/p^{\ominus})$

$$\Delta_r G = \mu_B - \mu_A = \left\{\mu_B^{\ominus} + RT\ln\left(\frac{p_B}{p_B^{\ominus}}\right)\right\} - \left\{\mu_A^{\ominus} + RT\ln\left(\frac{p_A}{p_A^{\ominus}}\right)\right\}$$

Hence, using $\Delta_r G^{\ominus} = \mu_B^{\ominus} - \mu_A^{\ominus}$,

$$\Delta_r G = \Delta_r G^{\ominus} + RT\ln\left(\frac{p_B}{p_A}\right) = \Delta_r G^{\ominus} + RT\ln Q$$

Equilibrium Constant

$$\Delta_r G = \Delta_r G^\ominus + RT \ln Q$$

At equilibrium, $\Delta_r G = 0$

$$0 = \Delta_r G^\ominus + RT \ln Q_{eq} = \Delta_r G^\ominus + RT \ln K$$

Hence,

$$\Delta_r G^\ominus = -RT \ln K$$, where $K = (p_B/p_A)_{eq}$

Generalization

For an arbitrary reaction,

$$\sum\nolimits_{\text{reactant } i} \nu_i \,\{i\} \rightleftarrows \sum\nolimits_{\text{product } j} \nu_j \,\{j\}$$

e.g. 2A + B ⇄ 3C + D

ν_{A} = 2, ν_{B} = 1, ν_{C} = 3, ν_{D} = 1

Now,

$$\mathrm{d}G = \sum\nolimits_{\text{product } j} \mu_j \mathrm{d}n_j + \sum\nolimits_{\text{reactant } i} \mu_i \mathrm{d}n_i$$

Generalization

$$\mathrm{d}G = \sum\nolimits_{\text{product } j} \mu_j \mathrm{d}n_j + \sum\nolimits_{\text{reactant } i} \mu_i \mathrm{d}n_i$$

$$\mathrm{d}G = \sum\nolimits_{\text{product } j} \mu_j \nu_j \mathrm{d}\xi - \sum\nolimits_{\text{reactant } i} \mu_i \nu_i \mathrm{d}\xi$$

$$\mathrm{d}G = \left(\sum\nolimits_{\text{product } j} \mu_j \nu_j - \sum\nolimits_{\text{reactant } i} \mu_i \nu_i\right) \mathrm{d}\xi$$

$$\Delta_\mathrm{r} G = \left(\frac{\partial G}{\partial \xi}\right)_{p,T} = \sum\nolimits_{\text{product } j} \mu_j \nu_j - \sum\nolimits_{\text{reactant } i} \mu_i \nu_i$$

Topic 6A

Chemical Potential and Activity

- Real gas: $\mu = \mu^{\ominus} + RT \ln(f/p^{\ominus})$
- Real solution (solvent): $\mu = \mu^{*} + RT \ln a$
- Real solution (solute): $\mu = \mu^{\ominus} + RT \ln a$

- Generally, let us use

$$\mu = \mu^{\ominus} + RT \ln a$$

Generalization

$$\Delta_{\mathrm{r}}G = \sum\nolimits_{\mathrm{product}\ j} \nu_j \mu_j - \sum\nolimits_{\mathrm{reactant}\ i} \nu_i \mu_i$$

For each species x,

$$\nu_x \mu_x = \nu_x(\mu_x^{\ominus} + RT \ln a_x) = \nu_x \mu_x^{\ominus} + RT \ln\left(a_x^{\nu_x}\right)$$

Then

$$\sum\nolimits_x \nu_x \mu_x = \sum\nolimits_x \nu_x \mu_x^{\ominus} + RT \sum\nolimits_x \ln\left(a_x^{\nu_x}\right)$$

$$= \sum\nolimits_x \nu_x \mu_x^{\ominus} + RT \ln \prod\nolimits_x a_x^{\nu_x}$$

Generalization

$$\Delta_r G = \sum_{\text{product } j} \nu_j \mu_j - \sum_{\text{reactant } i} \nu_i \mu_i$$

Substitution leads to

$$\Delta_r G = \Delta_r G^{\ominus} + RT \ln \prod_{\text{product } j} a_j^{\nu_j} - RT \ln \prod_{\text{reactant } i} a_i^{\nu_i}$$

where $\Delta_r G^{\ominus} = \sum_{\text{product } j} \nu_j \mu_j^{\ominus} - \sum_{\text{reactant } i} \nu_i \mu_i^{\ominus}$

$$\Delta_r G = \Delta_r G^{\ominus} + RT \ln \left(\prod_{\text{product } j} a_j^{\nu_j} \Big/ \prod_{\text{reactant } i} a_i^{\nu_i} \right)$$

Q and K

$$\Delta_\mathrm{r}G = \Delta_\mathrm{r}G^\ominus + RT\ln Q$$

where the reaction quotient Q is defined as

$$Q = \frac{\prod_{\mathrm{product}\ j} a_j^{\nu_j}}{\prod_{\mathrm{reactant}\ i} a_i^{\nu_i}}$$

At equilibrium, $\Delta_\mathrm{r}G = 0$

$$0 = \Delta_\mathrm{r}G^\ominus + RT\ln Q_\mathrm{eq} = \Delta_\mathrm{r}G^\ominus + RT\ln K$$

where the equilibrium constant K is Q at equilibrium

Connection to Thermochemistry

$$\Delta_\mathrm{r}G^{\ominus} = -RT\ln K$$

Since we know

$$\Delta_\mathrm{r}G^{\ominus} = \sum_{\text{products } j} \nu_j \Delta_\mathrm{f}G^{\ominus}(j) - \sum_{\text{reactants } i} \nu_i \Delta_\mathrm{f}G^{\ominus}(i)\,,$$

we can calculate K from thermochemical data.

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Meaning of Q and K

- Reaction quotient Q
  - "How far the equilibrium is"
  - Varies as the reaction progresses
  - $Q = K$ at equilibrium

- Equilibrium constant K
  - Amounts of reactants and products at equilibrium
  - Constant for the given experimental condition

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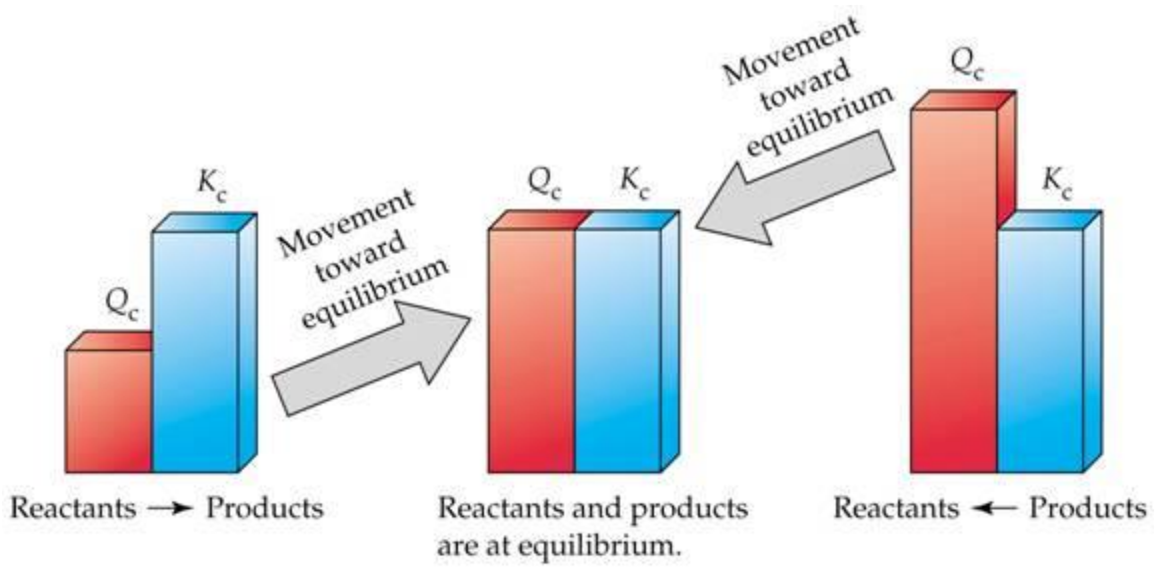
Meaning of Q and K



(Image: http://www.fccj.us/chm2046/SampleTest/46M10dAnswers.htm)